

Cyndi Silva Abanto

Bachelor in Industrial and Systems Engineering | AI & Data Science

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Bachelor in Industrial and Systems Engineering based in Peru, with a research focus on Deep Learning, Computer Vision, and Generative AI applied to spatiotemporal and scientific data. Author of a CONCYTEC-funded undergraduate thesis on radar-based precipitation nowcasting using deep neural networks, with findings presented at the LatinX in AI Workshop at CVPR 2026 and the WiCV Workshop at ECCV 2026. Experienced in Python and SQL for ETL pipelines and large-scale data processing, with hands-on work in Machine Learning, RAG systems, and anomaly/predictive modeling using PyTorch, TensorFlow, and Hugging Face on Microsoft Azure, OCI, and Linux/HPC environments. Actively engaged with the Latin American AI research community through PhawAI, TaReCDa, and the Summit of AI in LatAm, and motivated to keep building AI-driven solutions with regional and scientific impact.

Research interests: Deep Learning for spatiotemporal and scientific data, Computer Vision, Generative AI & RAG, anomaly detection.

Education

Universidad de Piura (UDEP)

Bachelor in Industrial and Systems Engineering

Piura, Peru

2020-II – 2025-II

TECH Global University

University Expert in Advanced Techniques in Computer Vision

Jan. 2026 – Jun. 2026

Research Recognition & LatAm AI Community

- Travel Grant — 2nd Workshop on Video Large Language Models, CVPR 2026, USA (Jun. 2026)
- LatinX in AI Workshop @ CVPR 2026, USA (2026): presented the poster “PiuraNet: Spatiotemporal Deep Learning for Radar-Based Precipitation Nowcasting in Piura.”
- Funded Research Project — CONCYTEC Undergraduate Thesis Competition, Peru (2025): selected for funding in a competitive national program.
- Summit of AI in LatAm — Sala’26, Ecuador (2026): presented projects in an academic poster session as part of an intensive deep learning program.
- PhawAI — Connecting Peruvian Researchers in AI, Peru (2025): scholar in AI training and research; travel grant recipient for on-site phase.
- TaReCDa Regional Workshop, Ecuador (2023): scholar in specialized applied data science training; travel grant recipient for on-site phase.
- Aspire Leaders Program, Peru (2025): developed an AI solution addressing a regional socioeconomic issue.

Certifications & Additional Training

- Oracle Cloud Infrastructure AI Foundations Associate (June 2026)
- Data Science — ONE Tech Foundation, Oracle Next Education & Alura Latam (Jan. 2025 – Mar. 2026)
- Introduction to Deep Learning — NVIDIA Deep Learning Institute (Feb. 2025)

Work Experience

Estación Científica Ramón Múgica — Piura, Peru

Research Assistant

Jan. 2026 – May 2026

- Processed, transformed, and analyzed large volumes of meteorological data using PostgreSQL and SQL Server, developing views, stored procedures, functions, and triggers to automate and optimize data processing; built ETL processes for large-scale data handling using Dask.
- Developed ETL pipelines for the integration, transformation, and storage of data from multiple weather stations, incorporating NOAA APIs and Microsoft Azure services, with data quality verification using Pydantic.
- Implemented Machine Learning models for prediction and anomaly detection, using pretrained Hugging Face models, Transformers for time series, and Isolation Forest.
- Developed interactive Power BI dashboards to monitor variables and temporal patterns, facilitating analysis and tracking of key indicators.

Universidad de Piura (UDEP) — Piura, Peru

Professional Operations Intern, Faculty of Engineering — Project Management Office

Jan. 2025 – Dec. 2025

- Supported the elicitation and analysis of functional requirements, use case definition, and process modeling through UML diagrams and flowcharts, as part of project formulation and management under PMI standards and agile methodologies (Scrum).
- Implemented interactive dashboards in Power BI and Microsoft Project to monitor project progress, manage risks, track indicators, and verify milestone completion.
- Developed synchronization and automation processes using Excel (VBA, advanced formulas, and Power Query) and Microsoft Power Platform (Power Apps and Power Automate), optimizing operational processes and reducing manual tasks.

Universidad de Piura (UDEP) — Piura, Peru

Mar. 2023 – Dec. 2024

Teaching Assistant — Database Systems and Linear Algebra

- Supported students in the design and implementation of relational databases, including modeling, SQL queries, variable interpretation, data import/export, and report generation in MySQL.
- Evaluated student academic performance and provided structured feedback to strengthen technical competencies; conducted workshops on linear algebra fundamentals, basic programming, and database management.
- Managed course academic records, preparing progress reports and supporting the organization and oversight of academic activities.

Research & AI Projects

Rainfall Prediction with Deep Neural Networks on Radar Data — Piura, Peru

Jun. 2025 – Apr.

2026

- Developed and evaluated Deep Learning models (U-Net, Swin Transformer, ConvLSTM, cGAN, and diffusion models) for precipitation nowcasting, processing multidimensional radar data and analyzing spatial and temporal patterns through classification, regression, and structural similarity metrics.
- Optimized training using class imbalance handling techniques and loss functions, improving the detection of extreme events and anomalous patterns in spatiotemporal data; implemented MLflow for experiment tracking, metrics, and model versioning, ensuring traceability and reproducibility.

BioSpace — NASA Space Apps Challenge

Oct. 2025

- Developed a platform for classifying space biology scientific articles using Transformer models (SciBERT and BiomedBERT) from Hugging Face, deployed on Microsoft Azure and backed by a MongoDB database.
- Implemented a Retrieval-Augmented Generation (RAG) system that used retrieved documents to enrich context and generate more accurate, grounded responses, evaluating performance with metrics such as Cosine Similarity and BERTScore.
- Developed interactive frontend features with JavaScript, collaborating within a multidisciplinary team that contributed to the solution earning national ranking and an honorable mention.

Social Engineering Fraud Detection with Deep Learning

Feb. 2025

- Designed an end-to-end system for anomaly and social engineering fraud detection integrating multivariate datasets; applied ADASYN to address the severe class imbalance typical in fraud data.
- Evaluated classical and Deep Learning models, optimizing Recall and Precision metrics ($AUC \approx 85\%$) to reduce false positives and improve the detection of malicious transactions/messages.

LLMs for Legal Documents — Piura Judicial System

Aug. 2023 – Dec. 2023

- Developed a Generative AI system based on Llama 3.2 to transform colloquial-language queries into formal legal terminology, integrating NLP techniques and Bag-of-Words representation for query processing and analysis.
- Integrated REST APIs for communication between the application and the database, automating the querying, storage, and management of legal documents.
- Trained and evaluated language models, achieving an initial performance of 58%, establishing a baseline for continuous system improvement.

Skills

- **Programming Languages:** Python, C++, JavaScript, HTML, CSS
- **Data Science & Machine Learning:** Machine Learning, Deep Learning, anomaly detection, classification, regression, time series, EDA, Scikit-learn, Pandas, NumPy, OpenCV, PyTorch, TensorFlow, Hugging Face
- **Databases & Cloud:** SQL (PostgreSQL, SQL Server), Autonomous Database, Git, Data Pipelines (ETL), Oracle Cloud Infrastructure (OCI), Microsoft Azure
- **Advanced AI:** PyTorch, TensorFlow, Hugging Face, HPC (Slurm), CUDA/GPU, AI agents
- **Languages:** Spanish (native), English (intermediate, B1)